

1. Administrative & Study Identification

Full Study Title: An Open-label Study of Encorafenib + Binimetinib in Patients With BRAFV600-mutant Non-small Cell Lung Cancer **Short Title/Acronym:** PHAROS **Protocol Number:** C4221008 (ARRAY-818-202) **NCT Number:** NCT03915951 **Posting ID:** 99482934891306977 **Sponsor Name:** Pfizer (Array BioPharma Inc.) **Trial Status:** COMPLETED (Study has been completed) **Investigational Product:** Encorafenib (BRAFTOVI®) and Binimetinib (MEKTOVI®) **Phase of Development:** Phase II **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the efficacy, specifically the Objective Response Rate (ORR), of encorafenib given in combination with binimetinib in patients with BRAF V600E-mutant metastatic non-small cell lung cancer (NSCLC). **Secondary Objectives:** To evaluate the Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), Disease Control Rate (DCR), and the overall safety and tolerability of the combination.

2.2 Background & Rationale Lung cancer is the leading cause of cancer-related mortality. BRAF V600E mutations occur in approximately 1% to 2% of patients with NSCLC and define a biologically distinct subset of the disease. Dual inhibition of the MAPK pathway with BRAF and MEK inhibitors is currently recommended by clinical guidelines as the preferred first-line approach for patients with BRAF V600E-mutant metastatic NSCLC. This study aims to evaluate the combination of encorafenib and binimetinib as a targeted therapeutic approach for this specific genetic subpopulation.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 98 patients **Number of Sites:** Multicenter (United States, e.g., Emory University Hospital, Johns Hopkins University) **Estimated Study Start:** 2019 **Estimated Primary Completion:** Primary analyses conducted as of September 2022

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically confirmed diagnosis of non-small cell lung cancer (NSCLC) that is currently Stage IV. 2. Presence of a BRAF V600E mutation in lung cancer tissue as determined by an FDA-approved or local laboratory assay. 3. Patients must be treatment-naïve, or have received prior systemic therapy for metastatic disease (e.g., first-line standard platinum-based chemotherapy or an anti-PD-1/PD-L1 inhibitor).

4.2 Exclusion Criteria 1. Prior treatment with any BRAF or MEK inhibitors. 2. Patients with symptomatic brain metastasis, leptomeningeal disease, or other active central nervous system (CNS) metastases. 3. History or current evidence of retinal vein occlusion (RVO) or current risk factors for RVO (e.g., uncontrolled glaucoma).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Oral Encorafenib 450 mg once daily plus oral Binimetinib 45 mg twice daily. **Route of Administration:** Oral **Duration of Treatment:** Administered in continuous 28-day cycles until disease progression, unacceptable toxicity, or treatment discontinuation. **ATC Code:** L01EE03 (Binimetinib) **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/mektovi-epar-product-information_en.pdf

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease symptoms or treatment-related toxicities (e.g., antiemetics for nausea). **Prohibited:** Prior or concurrent treatment with other BRAF or MEK inhibitors.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, BRAF V600E Mutation Testing, Baseline Imaging
Baseline	Day 0	First Dose, Vital Signs, Safety Labs
Treatment Period	Every 28 days (per cycle)	Dispense Medication, Safety Labs, Efficacy Assessment (Tumor Imaging per RECIST v1.1)
End of Study	Upon progression/toxicity	Final Vital Signs, Exit Interview, IP Accountability, Long-Term Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR), defined as the percentage of patients with a confirmed complete or partial response. **Measurement Method:** Independent Radiology

Review (IRR) using Response Evaluation Criteria in Solid Tumors (RECIST) v1.1.

7.2 Secondary & Exploratory Endpoints * Duration of Response (DOR) * Progression-Free Survival (PFS) * Overall Survival (OS) * Disease Control Rate (DCR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection during scheduled visits. Common Treatment-Related Adverse Events (TRAEs) monitored include nausea, diarrhea, fatigue, and visual changes. **Serious Adverse Events (SAEs):** Routine reporting timeline, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent DMC utilized to review ongoing safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy evaluations were pre-specified and conducted separately for treatment-naïve and previously treated patient cohorts. Safety set includes all patients who receive at least one dose of study medication. **Sample Size Justification:** Sized to provide adequate power to evaluate the primary endpoint (ORR) in both cohorts with precision. **Handling of Missing Data:** Efficacy data like PFS and OS are analyzed utilizing standard time-to-event methodologies (e.g., Kaplan-Meier curves); unreached events or missing data are appropriately right-censored.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure data integrity and patient safety according to the sponsor's clinical protocol. **Audit Trail:** Robust tracking mechanisms, eCRF locking processes, and data clarification forms via standard quality audit systems.

11. Study Identifiers & Governance

Primary ID: C4221008 **Registry Number:** NCT03915951 **Posting ID:** 99482934891306977 **Sponsor/Lead Organization:** Pfizer **Study Title:** PHAROS **Official Regulatory Title:** A Phase 2, Open-label Study of Encorafenib + Binimetinib in Patients With BRAFV600-mutant Non-small Cell Lung Cancer **Source Level:** 5

12. Clinical Framework (Oncology/NSCLC Specific)

Primary Condition: BRAF V600E-Mutant Metastatic Non-Small Cell Lung Cancer (mNSCLC) **Diagnosis Threshold:** Confirmed Stage IV NSCLC possessing a BRAF V600E mutation via FDA-approved testing **Medical Methodology:** Targeted combination kinase inhibition (BRAF and MEK inhibitors) **Optimization Target:** Maximizing Objective Response Rate (ORR) and delaying disease progression

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for first- and second-line metastatic NSCLC **Education Protocol (HCP):** Site-level training on identifying and proactively managing targeted therapy toxicities **Education Protocol (Patient):** Onsite training regarding daily oral medication adherence and AE self-reporting (e.g., reporting visual disturbances or severe GI symptoms) **Quality Audit Mechanism:** Independent Radiology Review (IRR) for unbiased confirmation of tumor responses

14. Observation & Follow-Up Schedule

Total Duration: Continuous until disease progression or unacceptable toxicity, followed by long-term survival follow-up **Onsite Visit Cadence:** Start of every 28-day cycle **Remote Monitoring Frequency:** Periodic check-ins for AE tracking **Checklist Review Interval:** End of each 28-day cycle

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator			[Insert Name]			_____		[DD-MMM-YYYY]								
	Clinical Operations Lead			[Insert Name]			_____		[DD-MMM-YYYY]								
	Lead Statistician			[Insert Name]			_____		[DD-MMM-YYYY]								